Theoretical Computer Science Exam, September 6, 2024

The exam consists of **<u>4 exercises</u>**. Available time: 1 hr 30 min.; you may write your answers in Italian, if you wish.

All exercises must be addressed for the written exam to be considered sufficient.

F.NAME …………………………….. L.NAME …………………………………… PERSON CODE …………………

Exercise 1 (8 pts)

Consider the following language

$$L = \{ a^i b^j c^k d \mid j > i + k \}$$

For instance $abbbcd \in L$, $aabbcd \notin L$.

1. Design a minimal power automaton that accepts language L. Write a computation of the automaton that accepts string *abbbcd* and one that rejects string *aabbcd*.
2. Design a grammar, of the least general type, that generates language L. Write a derivation in the grammar for the string $abbbcd \in L$.

Exercise 2 (8 pts)

a) Specify, by means of a sentence of Monadic First-Order logic (or Second-Order only if necessary) the language L_1 over the alphabet $\{a, b, c\}$ defined by the following condition. All pairs of nonconsecutive letters a have a letter c between them, and there is a pair of nonconsecutive letters c with no letter a between them. For instance, $babcbabcacbbcb \in L_1$, while $babbacbabcb \notin L_1$.

b) Specify, by means of a sentence of Monadic First-Order logic (or Second-Order only if necessary) the language L_2 over the alphabet $\{a, b\}$ defined by the following condition. There is at least one occurrence of letter b at an even position, and at least one occurrence of letter b at an odd position. For instance, $abba \in L_2$, while $abab \notin L_2$.

Exercise 3 (8 pts)

Say if the following sets are decidable and if they are semidecidable; provide suitable, precise justifications for your answers.

1. $S_1 = \{ n \mid \exists z$ such that the Turing machine M_z accepts all strings of length $n \}$
2. $S_2 = \{ n \mid \forall z$ the Turing machine M_z accepts some string of length $n \}$
3. $S_3 = \{ z \mid \forall n$ the Turing machine M_z does not accept any string of length $n \}$
4. $S_4 = \{ z \mid \exists n$ such that the Turing machine M_z accepts some string of length $n \}$

Exercise 4 (8 pts)

Consider again the language L of Exercise 1

$$L = \{ a^i b^j c^k d \mid j > i + k \}$$

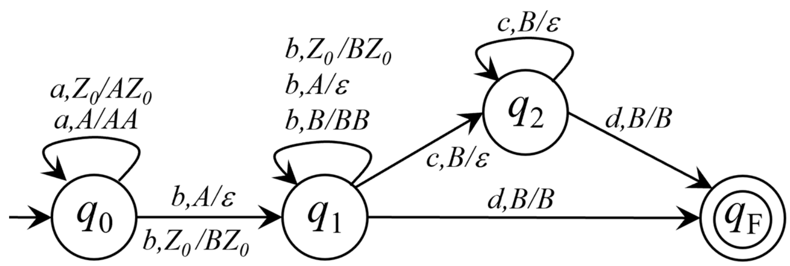
1. Describe in a precise, unambiguous way, a 1-tape Turing machine (i.e., having 1 memory tape) that recognizes language L. Determine its time and space complexity class as a function of the length n of the input string.
2. Describe in a precise, unambiguous way, a RAM machine that accepts language L, and determine its time and space complexity class, as a function of the length n of the input string, according to both the constant and the logarithmic cost criterion.

Solutions

Exercise 1

1. Design a minimal power automaton that accepts language L. Write a computation of the automaton that accepts string *abbbcd* and one that rejects string *aabbcd*.

A deterministic push down automaton suffices.



$< abbbcd, q_0, Z_0 > \vdash\ <bbbcd, q_0, AZ_0> \vdash\ <bbcd, q_1, Z_0> \vdash <bcd, q_1, BZ_0> \vdash <cd, q_1, BBZ_0> \vdash <d, q_2, BZ_0>$

$\vdash <\varepsilon, q_F, BZ_0>$ accept

$< aabbcd, q_0, Z_0 > \vdash\ <abbcd, q_0, AZ_0> \vdash\ <bbcd, q_0, AAZ_0> \vdash <bcd, q_1, AZ_0> \vdash <cd, q_1, Z_0>$ reject

2. Design a grammar, of the least general type, that generates language L. Write a derivation in the grammar for the string $abbbcd \in L$.

A context-free grammar suffices. Axiom is S.

$S \rightarrow A\ B\ C\ d$

$A \rightarrow a\ A\ b \ \mid\ \varepsilon$

$B \rightarrow b\ B \mid\ b$

$C \rightarrow b\ C\ c \ \mid\ \varepsilon$

$S \Rightarrow A\ B\ C\ d \Rightarrow a\ A\ b\ B\ C\ d \Rightarrow a\ b\ B\ b\ C\ c\ d \Rightarrow a\ b\ B\ b\ c\ d \Rightarrow a\ b\ b\ b\ c\ d$

Exercise 2

$\forall x\ \forall y\ (\ y > x+1\ \wedge\ a(x) \wedge\ a(y)\ \rightarrow\ \exists z\ (\ x < z < y \wedge c(z)\ )\ )\ \wedge$

$\exists x \exists y\ (\ y > x + 1\ \wedge\ c(x)\ \wedge\ c(y)\ \wedge\ \forall z\ (\ x < z < y\ \rightarrow \neg\ a(z)\ )\ )$

$\exists\ X\ \forall x\ (\ X(0) \wedge$

$(\neg\ last(x)\ \rightarrow\ (\ X(x+1)\ \leftrightarrow\ \neg\ X(x)\ )\ )\ \wedge$

$\exists y\ (\ X(y)\ \wedge b(y)\ )\ \wedge$

$\exists z\ (\ \neg X(z)\ \wedge b(z)\ )$

$)$

Exercise 3

1. $S_1 = \{\ n \mid \exists z$ such that the Turing machine M_z accepts all strings of length $n\ \}$

$S_1 = \mathrm{N}$ (the set of all natural numbers) because for every possible length n there are Turing machines that accept all strings of that length. Hence S_1 is decidable and semidecidable.

2. $S_2 = \{\ n \mid \forall z$ the Turing machine M_z accepts some string of length $n\ \}$

$S_2 = \varnothing$ because for no possible length n all Turing machines accept some string of that length. For instance Turing machines accepting the empty language do not accept any string of any length. Hence S_2 is decidable and semidecidable.

3. $S_3 = \{\ z \mid \forall n$ the Turing machine M_z does not accept any string of length $n\ \}$

S_3 is the set of (indices of) Turing machines that accept the empty language; it satisfies the hypothesis of the Rice theorem, hence S_3 is undecidable. It is not semidecidable because its complement, i.e., the set of Turing machines that accept a nonempty language, is semidecidable (by means of dovetail simulation).

4. $S_4 = \{\ z \mid \exists n$ such that the Turing machine M_z accepts some string of length $n\ \}$

S_4 is the set of (indices of) Turing machines that accept a non-empty language; it satisfies the hypothesis of the Rice theorem, hence S_4 is undecidable. It is semidecidable by means of dovetail simulation.

Exercise 4

1. Describe in a precise, unambiguous way, a 1-tape Turing machine (i.e., having 1 memory tape) that recognizes language L. Determine its time and space complexity as a function of the length n of the input string.

The 1-tape Turing machine uses the memory tape as a stack: it reads the a's and counts them in unary in the memory tape. Then it reads the b's canceling a symbol in the tape for every b it reads until the tape becomes empty. Next it stores one symbol in the tape for every successive b and cancels one symbol for every c. It accepts the string iff the tape is not empty when the final d is read. The time complexity is $\Theta(n)$ and the space complexity is also $\Theta(n)$, as the worst case occurs when the input string is composed only of a symbols.

2. Describe in a precise, unambiguous way, a RAM machine that accepts language L, and determine its time and space complexity class according to both the constant and the logarithmic criterion.

The RAM machine reads the string and counts the number of a's, b's and c's using three variables, then it adds the numbers of a's and c's and compares the sum with the number of b's.

Adopting the uniform criterion, the space complexity is constant, and the time complexity is $\Theta(n)$.

Adopting the logarithmic criterion, the space complexity is $\Theta(\log n)$ and the time complexity is $\Theta(n \log n)$.